

## 21-128 and 15-151 Problem Sheet 2

Solutions to the following six exercises and optional bonus problem are to be submitted to Gradescope by

11 PM on Wednesday, the 9th of September, 2026.

---

**Instructions.** This assignment has 30 points, together with an optional 1-point bonus problem. Unless a problem explicitly says otherwise, every answer must be justified by a proof. A counterexample must satisfy all of the hypotheses and must be checked explicitly. We use  $\subseteq$  for containment and  $\subsetneq$  for proper containment.

### Problem 1

[4 points]

For each statement below, determine whether it is true for every choice of nonempty sets  $A, B, C \subseteq \mathbb{Z}$ . Prove each statement that is always true, and give a fully specified counterexample to each statement that is not always true.

(a)

$$A \cap B \cap C \subsetneq A \cup B.$$

(b)

$$(A \setminus B) \cap (B \setminus A) = \emptyset.$$

(c)

$$(A \cap B \neq \emptyset) \implies (A \setminus B \subsetneq A).$$

### Problem 2

[5 points]

For each  $n \in \mathbb{Z}_{>0}$ , define

$$M_n = \{x \in \mathbb{Z} : n \mid x\}.$$

Let  $m, n \in \mathbb{Z}_{>0}$ .

(a) Prove that

$$M_m \subseteq M_n \iff n \mid m.$$

(b) Prove that

$$M_m = M_n \iff m = n.$$

### Problem 3

[4 points]

For sets  $X$  and  $Y$ , their *symmetric difference* is

$$X \triangle Y = (X \setminus Y) \cup (Y \setminus X).$$

Prove that, for all sets  $A, B, C$ ,

$$C \setminus (A \triangle B) = (A \cap B \cap C) \cup (C \setminus (A \cup B)).$$

Your proof must proceed by showing that an arbitrary element belongs to the left-hand side if and only if it belongs to the right-hand side. A Venn diagram may be used for discovery, but it is not a proof.

#### Problem 4

[5 points]

A set  $T$  is called *transitive* if, for all sets  $x$  and  $y$ ,

$$(x \in y \wedge y \in T) \implies x \in T.$$

(a) Determine, with proof, which of the following sets are transitive:

$$T_1 = \{\emptyset, \{\emptyset\}\}, \quad T_2 = \{\{\emptyset\}\}.$$

(b) Prove that if  $T$  is transitive, then its power set  $\mathcal{P}(T)$  is also transitive. Your proof should explicitly distinguish membership from containment.

#### Problem 5

[5 points]

A set  $S \subseteq \mathbb{R}$  is called *open* if

$$(\forall x \in S)(\exists \varepsilon > 0)(\forall y \in \mathbb{R})(|x - y| < \varepsilon \implies y \in S).$$

Let  $S \subseteq \mathbb{R}$  be open, and suppose that  $a, b \in S$  with  $a < b$ . Prove that  $S \cap (a, b)$  contains infinitely many distinct points. Your proof should explicitly construct such points from an  $\varepsilon$  supplied by the definition of openness.

#### Problem 6

[7 points]

Define

$$f : \mathbb{Z}_{>0} \times \mathbb{Z}_{>0} \longrightarrow \mathbb{R}, \quad f(a, b) = \frac{(a+1)(a+2b)}{2}.$$

(a) Prove that the image of  $f$  is contained in  $\mathbb{Z}_{>0}$ .

(b) Determine

$$\text{im}(f) \cap \{1, 2, \dots, 20\},$$

and use your answer to state a conjecture describing the entire image of  $f$ .

(c) Prove your conjecture: determine exactly which positive integers belong to the image of  $f$ .

*Hint for part (c).* Set  $u = a + 1$  and  $v = a + 2b$ . Observe that  $2f(a, b) = uv$ , that  $2 \leq u < v$ , and that  $u$  and  $v$  have opposite parity. For the reverse direction, it may be useful to write a positive integer as  $2^k q$  with  $q$  odd.

**Bonus Problem**

[1 bonus point]

A function  $u : \mathbb{R} \rightarrow \mathbb{R}$  is *even* if  $u(-x) = u(x)$  for every  $x \in \mathbb{R}$ , and is *odd* if  $u(-x) = -u(x)$  for every  $x \in \mathbb{R}$ . Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be an arbitrary function.

- (a) Prove that there exists a unique pair of functions  $g, h : \mathbb{R} \rightarrow \mathbb{R}$  such that  $g$  is even,  $h$  is odd, and

$$f = g + h.$$

- (b) Suppose that

$$f(x) = c_0 + c_1x + \cdots + c_nx^n$$

is a polynomial function. Express the functions  $g$  and  $h$  from part (a) directly in terms of the coefficients  $c_0, c_1, \dots, c_n$ , and prove that your formulas work.